



Birla Institute of Technology & Science, Pilani

Pilani | Dubai | Goa | Hyderabad

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

First Semester 2026-27

Course Handout Part II

Date: 1 August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number	:[ME F315]
Course Title	:[Advanced Manufacturing Processes]
Instructor-in-charge	:[Tufan Chandra Bera]
[Team of Instructors]	:[Amit Kumar (T), amit.kumar@pilani.bits-pilani.ac.in , 2220-J] :[Pardeep Kumar (P), p20240464@pilani.bits-pilani.ac.in] :[Aashutosh Kumar (P), p20240493@pilani.bits-pilani.ac.in] :[Pradeep Shukla (P), p20220066@pilani.bits-pilani.ac.in] :[Pronamika Borthakur (P), p20220044@pilani.bits-pilani.ac.in]

1. Course description:

Unconventional manufacturing processes and analysis, micro-manufacturing technologies, introduction to computer-aided manufacturing, CNC machines, CNC Part programming, additive manufacturing, modelling and simulation of manufacturing processes, surface technologies, related laboratory exercise and fabrication project.

2. Scope and Objective:

This course introduces students to advanced manufacturing technologies for fabricating components across multiple length scales. It focuses on the challenges of machining difficult-to-machine materials and transforming them into high-value products. The course emphasizes hands-on learning through CNC part programming, CNC machine operation, additive manufacturing, and digital-to-physical manufacturing workflows. Lectures complement laboratory sessions by covering the fundamentals, process capabilities, numerical methods, and optimization techniques required for the analysis and development of advanced manufacturing processes.

3. Course learning outcomes:

CLO1. Explain the principles, capabilities, limitations, and applications of advanced manufacturing processes, including CNC machining, additive manufacturing, and non-traditional machining.

CLO2. Develop CNC part programs and implement digital manufacturing workflows to fabricate engineering components using advanced manufacturing systems.

CLO3. Analyze the influence of process parameters, material properties, and machine characteristics on product quality, productivity, and process performance.

4. Text books and references:

TB1. S. Kalpakjian and S. R. Schmid, “Manufacturing Engineering and Technology,” Pearson Education, 2001, New Delhi.

TB2. A. Ghosh and A. K. Mallik, “Manufacturing Science”, 2nd ed., Affiliated East-West Press, New Delhi, 2010.

RB1. G. F. Benedict, “Non Traditional Manufacturing Processes”, CRC Press, 2019.

RB2. V. K. Jain, “Advanced Machining Processes”, 2nd ed., Allied Publisher Bombay, 2010.

RB3. V. K. Jain (Editor), “Micromanufacturing Processes”, 1st ed., CRC Press, 2012.

RB4. M. Groover and E. Zimmers, “CAD/CAM: Computer-Aided Design and Manufacturing”, 1st ed., Pearson Education, 2003, New Delhi.

RB5. Gibson, D. W. Rosen and B. Stucker, “Additive manufacturing technologies: rapid prototyping to direct digital manufacturing”, Springer, 2010.

5. Course Format & Weekly Workflow

Wk no.	Module no.	Description	Ref.	Learning Outcome
1-3	M1: Unconventional Manufacturing processes (UMP) and analysis	L (1-8): Working principles, process parameters and analysis of various unconventional machining, forming, welding, and casting processes.	TB1, TB2, RB1 and RB2	To understand the basic concepts, working principles and analyze UMP
4-5	M2: Micro-manufacturing technologies	L (9-14): Scaling laws, micro-machining, micro-forming and micro-molding, micro-joining and assembly, micro-additive processes, micro-metrology and nanofabrication techniques.	RB3	To get acquainted with various micro-manufacturing technologies used in industries.
5-8	M3: Computer-Aided Manufacturing	L(15-16): Introduction of CAM, CNC machines,	TB1 and RB4	To learn fundamentals, CNC systems and hardware of CNC machines.
9-10	M4: CNC Part Programming	L (17-19): CNC Basic and advanced programming	RB4	To develop CNC programming and operation skill in CNC machines.
11-12	M5: Additive manufacturing	L (20-23): Basic principles and	TB1 and	To get exposure to the principles and working

	(AM)	working of different AM processes	RB5	of AM processes
13	M6: Modelling and simulation	L (24-26): Introduction to different modelling and simulation tools for manufacturing processes, understanding the modelling of different manufacturing processes and analyzing simulation results	RB4	To get exposure to different tools for modelling and simulation of manufacturing processes
14	M7: Surface technologies	L (27-28): Basics of tribology, lubrication, wear, cost of friction and wear	TB1 and TB2	To get acquainted with the fundamentals of the surface behavior of materials

6. Evaluation Scheme:

Component	Duration (Min.)	Weightage (%)	Date & Time	Nature of component (Closed Book/ Open Book)
Midsem Test	90 Min.	25	07/10/2026 FN1	Open Book
Comprehensive Examination	180 Min.	35	10/12/2026 FN	Closed Book
Tutorial	---	20	---	Open Book
Practical	---	20	---	Closed Book

Criterion for NC: The minimum passing marks of this course will be determined based on the 40% of the average score of the class strength. This criterion will be considered as passing marks for valid grade in this course.

Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

7. Chamber consultation hours: TBA

8. Makeup Policy: [Makeup will be given only for genuine cases (accompanied by proof) for Mid-Sem or comprehensive exam only.]

[Tufan Chandra Bera]
Instructor-in-charge
[ME F315]